# A Multiscale Cascaded Cross-Attention Hierarchical Network for Change Detection on Bitemporal Remote Sensing Images

Xiaofeng Zhang, Liejun Wang, and Shuli Cheng

***Abstract*— Remote sensing image change detection (RSCD) is an important task in remote sensing image interpretation. Some recent RSCD works focus on the extraction and interaction of global and local information; however, the current work underuses hierarchical features and may introduce noise from shallow encoders. In this article, we propose a multiscale cascaded cross-attention hierarchical network (MSCCA-Net). This network uses a large kernel convolution formed by stacking small kernel convolutions combined with an efficient transformer as the backbone network to achieve local and global feature extraction and fusion. We proposed for the first time the idea of bottom-up level-by-level fusion of hierarchical features, based on which we designed the multiscale cascade cross-attention (MSCCA) cross-fusion hierarchical features level by level from the bottom upward, realizing the redistribution of spatial and semantic information, and thus enhancing the gainful effect of the skip connection mechanism in the field of RSCD. Our experiments on three public datasets show that MSCCA is able to efficiently perform the reorganization of hierarchical features, thus avoiding misdetection and omission of small targets. Meanwhile, MSCCA-Net has more excellent comprehensive performance compared with other state-of-the-art methods.**

***Index Terms*— Change detection, convolutional neural network (CNN), multiscale cascaded cross-attention (MSCCA), remote sensing (RS), skip connection.**

## I. Introduction

THE rapid development of remote sensing (RS) image observation technology has provided a rich and reliable data source for the analysis of surface spatial information, which has further promoted the research of RS image change detection (RSCD). RSCD aims to identify changes in scenes of interest, such as changes in land cover or changes in the distribution of buildings, from RS images of the same location at different moments in time. The result of change detection is usually a generated binary change map, where the binary value indicates whether the area in which the pixel is located has changed [1]. The white areas in the change map shown in Fig. 1 indicate that a change has occurred, and the black areas indicate no change. RS change detection techniques play an important role in areas such as disaster damage assessment [2], urban development planning [3], and land management [4].

With many factors coming together, the search for more accurate and efficient change detection techniques remains a challenging task. First, coregistered RS images can vary in imaging conditions such as illumination, atmospheric disturbance, and observation angle depending on the time of acquisition. This can result in areas that have not changed appearing as different spectral and textural information in the RS image and thus being incorrectly classified as changed. Second, the failure to accurately align bitemporal RS images can confuse changed and unchanged areas, thus reducing the reliability of change detection results [5]. Changes that are not relevant to the application scene due to the above reasons are called "pseudo change." Finally, the complexity of information in high-resolution RS images increases the difficulty for the model to distinguish changes of interest from pseudo changes. The above problems are manifested in the change detection effect as false detection and missed detection, where the missed detection is mainly for small target areas. In addition, the problem of imbalance between changed and unchanged samples will also affect the performance of the model.

The CD task requires a pair of coregistered RS images as input. As shown in Fig. 1, there are three main structures that emerge from DL-based CD methods, depending on how the input image data is processed. As shown in Fig. 1(a), some CD methods stack the bitemporal images and input them into the network, called single-stream CD networks. FC-EF [1] directly performs feature extraction on the serially linked dual-temporal images used to detect changes. Obviously, due to the lack of independent original input images, it is difficult for the single-stream CD network to extract the fine bitemporal features of each of the dual-moment images. This does not facilitate the network to perform the subsequent variation detection process. As shown in Fig. 1(b), the dual-stream CD network uses two feature extraction branches with shared [4], [6], [7], [8], [9], [10], [11], [12] or nonshared [13], [14], [15] weights to obtain the respective feature features of the bitemporal image, which are then used as inputs to the change detection branch of the CD network after the corresponding feature fusion operation. Zhang et al. [16] extracted the features of each of the bitemporal images using a bitemporal

Manuscript received 10 October 2023; revised 21 January 2024; accepted 31 January 2024. Date of publication 2 February 2024; date of current version 14 February 2024. This work was supported in part by the Scientific and Technological Innovation 2030—Major Project under Grant 2022ZD0115800, in part by the Xinjiang Uygur Autonomous Region Tianshan Excellence Project under Grant 2022TSYCLJ0036, and in part by Postgraduate Scientific Research Innovation Project of Xinjiang Uygur Autonomous Region under Grant XJ2023G098. *(Corresponding authors: Liejun Wang; Shuli Cheng.)*

The authors are with the School of Computer Science and Technology, Xinjiang University, Ürümqi 830046, China (e-mail: zxf1332@stu.xju.edu.cn; wljxju@xju.edu.cn; cslxju@xju.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3361847

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

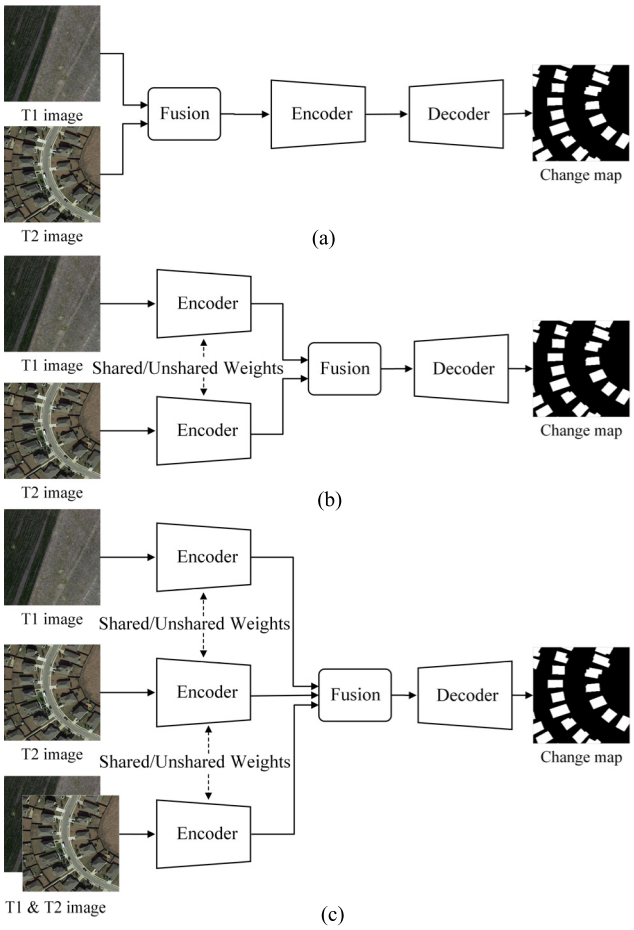


Fig. 1. Existing frameworks for change detection tasks. They can be classified into three categories based on the way in which the input image is processed, including single-stream methods, dual-stream methods, and triple-stream methods. (a) Single-stream CD network. (b) Dual-stream CD network. (c) Triple-stream CD network.

structure composed of full convolution. The features are then fed into a difference discrimination network for change detection to improve network performance. The shallow features extracted by the dual-stream CD network, however, carry too much noise, and due to the local connectivity of convolutional neural networks (CNNs), the extracted deep features lose contextual relevance. Based on the FC-Siam-Conc and FC-Siam-Di proposed by UNet [17], multiscale feature fusion using skip connection achieves the communication of deep and shallow information. As shown in Fig. 1(c), some recent works have attempted to combine the advantages of single-stream and dual-stream CD networks by inputting dual-moment images together with images of both in series or different images to a triple-stream CD network composed of dual-stream encoders and single-stream encoders. Zhao et al. [18] proposed a TSNet that uses the dual-moment images in series as the input to the third branch of the network, attempting to thoroughly extract and learn bitemporal change information through cross-stage learning, thus improving the network's ability to distinguish changes.

Usually, the hierarchical features extracted by the encoder supplement the upsampling process through skip connections; however, this skip connection is not always beneficial. A certain level of noise may be introduced when fusing the shallow features with the features of the final stage of the decoder via a skip connection. Since the final stage of the network usually consists of shallow convolutional or linear layers, these layers may not have sufficient nonlinear processing power to effectively remove the effects of this noise. This problem has been noted in some works, such as DMI-Net [9], combining self-attention and cross-attention [19], [20], [21] to control the feature distribution during feature fusion followed by a skip connection to suppress the interference of irrelevant change information; however, DMI-Net still lacks the interaction between features at different levels during the skip connection process. Such effective interaction of information can make fuller use of the semantic information of deep-level features and the spatial geometric information of shallow-level features, thus helping region localization and noise suppression. Based on the above considerations, we propose the multiscale cascaded cross-attention hierarchical network (MSCCA-Net) to address the noise effects of skip connection and the problems in the fusion and interaction of deep and shallow bitemporal features in the CD task. The network fuses the features extracted from each level of the twin encoder while performing bottom-up feature cross-attention so that each stage of the decoder can obtain sufficient information about the changing features. Our contributions are as follows.

1) We propose an end-to-end CD network based on a transformer and CNN. The network uses a large kernel convolution formed by stacking small kernel convolutions combined with an efficient transformer as the backbone network to achieve local and global feature extraction and fusion. The complementary advantages of the two can effectively eliminate the leakage and misdetection of small changes.
2) We design a multiscale cascade cross attention (MSCCA) module for the fusion of dual-branch features of CD network twin encoders. The bottom-up information flow in the fusion process enhances the use of deep semantic features, which can provide adequate feature representations for the decoding process and avoid the noise effects of shallow skip connections.
3) We conducted full experiments on three public benchmark datasets. The necessity of processing layered features before performing skip connection was elaborated in detail using quantitative and visual results in the experiment.

The rest of this article is organized as follows. Section II describes some related methods and self-attention mechanisms in the CD context. In Section III, our proposed framework is described in detail. Section IV presents a comprehensive experimental analysis. Section V discusses the validity of the modules and the mechanism of action of MSCCA. Finally, the conclusions of this article are drawn in Section VI.

## II. Related Work

### A. Traditional CD Methods

Before deep learning (DL) methods were applied to the field of CD, traditional algorithms had done a lot of research on the problem of change detection in RS images. Some traditional CD methods perform pixel-level calculations

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

directly on preprocessed RS images, such as difference, change vector analysis, and ratio methods [22]. Weismiller et al. [23] have used different methods to analyze change detection in coastal environments. Malila [24] and others have combined the magnitude and direction of the change vector to analyze whether and what changes have occurred. Such methods usually require a threshold to be set to distinguish between areas of change and areas of no change. Some CD methods are based on image transformation ideas, using feature representations extracted from RS images to distinguish between changed and unchanged areas. For example, Celik [25] applied principal component analysis to extract effective different features from multitemporal images by maximizing the variance of image differences to achieve change detection. There are also approaches [25], [26] that use clustering algorithms, such as $k$-means and fuzzy $c$-means, to categorize bitemporal image pixels into changed and unchanged classes. In addition, some traditional machine learning algorithms, such as Markov random fields [27], [28], decision trees [29], and support vector machines [30], [31] have also been used in change detection tasks.

Traditional methods mainly use manually extracted features relying on constructed unsupervised or supervised classification algorithms for change detection tasks. In the face of low-resolution and medium-resolution RS images, although the traditional methods can also achieve certain detection performance, their performance is too dependent on the threshold or classifier settings, and their generalization ability is insufficient. With the rapid development of RS technology, high resolution (HR) and very HR (VHR) RS images with richer texture and spectral information of ground objects are beginning to appear. In this case, it has been difficult for traditional CD methods to extract sufficient semantic information to meet the CD task requirements.

### B. Deep-Learning-Based CD Methods

In the face of today's massive amount of VHR RS images, DL algorithms, represented by CNN, have demonstrated their powerful potential for CD tasks [29], [30]. DL algorithms use hierarchical structures to extract low-level and high-level features from image data and identify the rich semantic information within them [31], [32]. DL-based CD methods thus avoid the tedious manual feature design phase of traditional CD algorithms based on different datasets and are, thus, able to handle a variety of complex RS image data.

Shallow features extracted by CNN contain rich image geometry and location information, while deep features contain more semantic features. The effective combination of deep features and shallow features can significantly improve the performance of CD networks. SNUNet [6], proposed based on the dense connectivity idea of UNet++ [32], uses long and short connections to fuse features at different scales, which reduces the semantic gap in the fusion of features at each level and reduces the loss of information in the feature of the landmarks. Several works have used dilated convolution [33] and the introduction of various forms of attention mechanisms [6], [16], [33], [34] in CD methods to enhance global information dependence and reduce noisy information interference. Among them, channel attention [6], [35] and spatial attention [34] focus on representative features by adaptively assigning different weights to the channel and spatial dimensions of the features, while Fang et al. [6] use channel attention to focus on bitemporal features extracted from different semantic levels. The inherent local connectivity property of CNN, however, leads to CNN-based CD methods not being able to fully use the attention mechanism for global modeling. The absence of global features results in the model being unable to consider the semantic details of the whole scene, which results in false and missed detections of small targets and uncertainty in changing edges.

### C. Self-Attention Mechanism

Self-attention [9], [36], [37] mechanisms have global attention capabilities and can help the network to capture long-range dependencies between contextual information to enhance feature representation. Inspired by the fields of natural language processing [38] and speech recognition [39], a vision transformer (ViT) [40] divides an image into multiple patches and then projects each patch as a fixed-length vector to feed the transformer for image classification tasks. In the CD domain, the transformer [41] has also demonstrated its power. Zhang et al. [7] designed the first pure transformer CD network, SwinSUNet, using a transformer module based on the shift window mechanism. ChangeFormer [8] combined a hierarchical twin transformer encoder with a lightweight MLP decoder and performed a combination that effectively obtains the multiscale remote details required for accurate CD. Compared to CNN, the transformer, however, lacks two inductive biases, local correlation and translation equivariance, which leads to the fact that the transformer may not perform as well as CNN methods, such as ResNet [42], when there is a lack of sufficient training data. Also, in the CD task, feeding two high-resolution RS images directly into the transformer module for processing inevitably leads to higher computational complexity and lower training efficiency. Many recent CD approaches have started to experiment with combining CNN and transformer to build networks that jointly capture global and local information for change detection tasks. BIT-CD [4] uses ResNet as the feature extractor and augments the features extracted by the convolutional network with a transformer. Wang et al. [43] proposed a network architecture called "UVACD" for bitemporal image change detection by combining a CNN for extracting semantic information with a visual transformer. MSTDSNet-CD [44] introduces deep supervision in a CD network constructed with a CNN and Swin transformer module for urban area change detection. The encoder part of ACABF-Net [14] consists of a CNN branch and a transformer branch in parallel, both of which extract the local and global information of RS images, respectively, and then fuse the local and global features by a bidirectional fusion method.

### D. Use of Multiscale Information

Under the challenges of image scale transformations and complex background environments, researchers are committed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

to mining multiscale information with a view to enhancing the superior performance of deep neural networks on CD tasks [45]. Multiscale feature maps contain varied spatial and semantic information, and their complementary nature provides strong support for the model to more accurately identify change regions and effectively reduce the interference caused by pseudo changes. At the same time, the fusion of multiscale features greatly enhances the accuracy of the model in the boundary detection of the change region and improves the localization accuracy of the change region and the robustness of the change detection. By appropriately fusing this multiscale information, the feature representation of change detection can be enriched, and the model's ability to recognize and segment the change region can be improved to enhance the detection effect.

In some studies, a pyramid structure is formed by stacking features of different scales on top of each other to achieve in-depth processing of the features. Lei et al. [46] proposed an innovative pyramid pooling module, which cleverly combines three convolution kernels of different scales to extract and integrate the deep features of an image. This approach can effectively capture multiscale features in RS images, providing richer information for image analysis and understanding. SLDDNet [47] introduces pyramid structure feature stacking, which establishes tight dependencies between local feature information through a central aggregation strategy. It adopts a phased approach at each level to fuse global semantic information and local detail information to maximize the use of multiscale feature information.

In addition, the attention mechanism plays a key role in the effective use of multiscale features. TSNet [18] uses a single-stream network to extract the depth features of the original image and uses a two-stream network to extract the different features of the dual-temporal phase RS image. In the feature fusion module, it uses a recurrent neural network and attention mechanism to achieve cross-phase feature integration, which significantly improves the network's ability to detect changes in dual-temporal phase RS images. USSFC-Net [13] employs multiscale decoupled convolution, which decouples convolution operations into spatial and channel correlation splicing to efficiently extract the multiscale features of the changed objects, while reducing the number of convolution parameters and computations. The spatial-spectral feature cooperation strategy proposed therein generates spatial-spectral cooperative 3-D weights through Gaussian modeling without adding any additional parameters, which effectively enhances the feature expression capability. ICIF-Net [48] extracts multiscale features in parallel through the CNN branch and transformer branch and extracts multiscale features in parallel through the intrascalar cross-interaction module at the same scale to obtain enhanced feature representation. It also performs feature fusion at different resolutions through both mask aggregation and spatial alignment. This approach achieves effective use of cross-scale features by extracting multiscale features in parallel and interacting them at the same scale, as well as fusing them at different resolutions, thus improving the performance of change detection.

A skip connection is a connection established between an encoder and a decoder that allows the decoder to access the feature map of the corresponding layer in the encoder. Through this mechanism, the decoder can use the high-level feature information learned by the encoder, thus allowing the network to effectively use the contextual information while maintaining the localization accuracy, which, in turn, leads to better image change detection results. In the TransUNetCD [5] model, the encoder is responsible for learning global contextual information, while the decoder focuses on recovering local details. Jump connections link high- and low-level feature maps, while the different enhancement module is used to generate enhanced difference features, and these strategies work together to achieve the task of change detection in RS images. Overall, the efficient extraction and use of multiscale information is a key challenge in the field of CD.

## III. Methodology

In this section, we first present the overall structure of the proposed method, then introduce the multibranch feature fusion (MFF) module and the MSCCA module. Finally, we introduce the loss function used for training the network.

### A. Architecture Overview

The architecture of MSCCA-Net is shown in Fig. 2. The overall network is a twin U-shape structure consisting of a weight-sharing dual-branch encoder and a single-branch decoder. Through skip connection, the hierarchical features extracted by the encoding branch are aggregated with the features at the same level of the decoding branch to make up for the loss of information, such as boundaries and locations, caused by downsampling. Unlike recent methods, the layered features extracted by the dual-branch encoder are fused and reorganized across scales by the MSCCA module during the skip connection process.

MSCCA-Net inputs 3-channel bitemporal RS images and outputs 2-channel change prediction maps. The bitemporal RS images used for change detection first pass through two layers of feature extraction module consisting of patch merging [18] and efficient transformer [49]. After reducing the spatial resolution to 1/4 of the original, the feature map is fed into a convolution block consisting of three $3 \times 3$ convolutions. As shown in Fig. 2, within the convolution blocks, we extracted features at different levels. When the input feature is $X_i \in \mathbb{R}^{(H/2^i)\times(W/2^i)\times C_i}$, the equivalent of features $\psi_3(X_i)$, $\psi_5(X_i)$, and $\psi_7(X_i)$ extracted by the $3 \times 3$, $5 \times 5$, and $7 \times 7$ convolution kernels can be obtained by stacking different levels of the convolution block. We then model the multiscale feature representation using the MFF module. This process is repeated three times in the encoding stage. In order to better recover the information lost during feature extraction through skip connection, the MSCCA module fuses the hierarchical features extracted by the dual-branch encoder while performing bottom-up cross-scale information reorganization. Each stage of the single-branch decoder consists of a patch expanding block and an efficient transformer, which restores the features to have the same resolution as the input as a result of change detection by upsampling them layer by layer.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

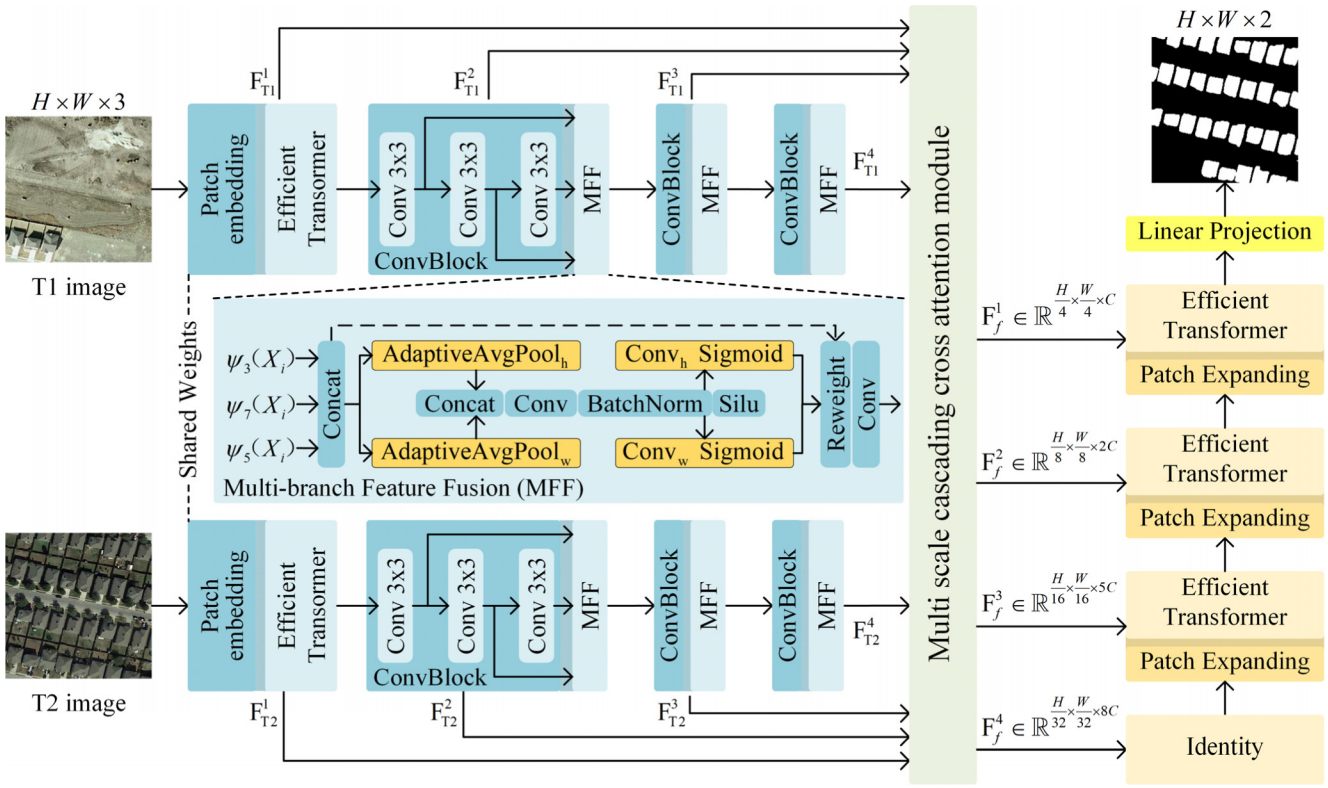


Fig. 2. Overall structure of the proposed MSCCA-Net.

### B. Multibranch Feature Fusion Module

The main structure of the MFF module is shown in Fig. 2. MFF first concatenates the input features $\psi_3(X_i)$, $\psi_5(X_i)$, $\psi_7(X_i) \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times C_i}$ in the channel dimension as $X_{\text{concat}} \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times 3C_i}$. Then, global pooling operations are performed on the width and height axes of the spliced features using adaptive average pooling, respectively, in a way that can be expressed as:

$$y^h = \frac{1}{W}\sum_{0\le i<W} x(h,i), \quad y^w = \frac{1}{H}\sum_{0\le j<H} x(j,w) \tag{1}$$

where $x$ denotes the single-channel form of the input feature, $y^h$ denotes the result of the pooling operation at height $h$ with kernel size $(1, W)$, and $y^w$ denotes the result of the pooling operation at height $w$ with kernel size $(H, 1)$. The use of biaxial feature pooling encoding to model each channel of the feature map at height and width can alleviate the loss of position information caused by traditional global flat pooling. We then perform spatial dimension splicing of the obtained biaxial features and channel adjustment from $nC$ to $nC/r$ using $1 \times 1$ convolution. The specific representation is as follows:

$$\begin{aligned} f' &= \text{Conv}_{1\times1}\big(\text{Concat}\big(\big[y^h, y^w\big]\big)\big) \\ f &= \text{SiLU}\big(\text{BatchNorm}\big(f'\big)\big) \end{aligned} \tag{2}$$

where $f \in \mathbb{R}^{(C/r)\times(H+W)}$ is a feature map with position information in both height and width directions and $r$ is the channel reduction factor, SiLU is the activation function used defined as:

$$\begin{aligned} \text{SiLU}(x) &= x\cdot\sigma(x) \\ \sigma(x) &= \frac{1}{1+e^{-x}}. \end{aligned} \tag{3}$$

Then, $f$ is divided into two tensors along the spatial dimension $f^h \in \mathbb{R}^{(C/r)\times(H)}$, $f^w \in \mathbb{R}^{(C/r)\times(W)}$. The two are transformed by $1 \times 1$ convolution and Sigmoid to adjust the channels and then reweight the feature $X_{\text{concat}}$ as the attention weights to achieve the channel attention

$$z^h = \text{sigmod}\big(\text{Conv}_{1\times1}\big(f^h\big)\big), \quad z^w = \text{sigmoid}(\text{Conv}_{1\times1}(f^w)) \tag{4}$$

$$Y = \text{Conv}_{1\times1}\big(X_{\text{concat}} \times z^h \times z^w\big). \tag{5}$$

The MFF module computes the attention weights using the spatial relationships of the input features $\psi_3(X_i)$, $\psi_5(X_i)$, $\psi_7(X_i)$, re-emphasizing the importance of channel information between multiscale features. Biaxial feature pooling coding is used in this process to reduce the positional errors in the subsequent coding process.

### C. Multiscale Cascaded Cross-Attention Module

The detailed structure of the MSCCA module is shown in Fig. 3. As a whole, the MSCCA module mainly fuses the hierarchical multiscale features extracted by the dual-branch encoder for the subsequent decoding process. Unlike the fusion approaches proposed by other networks, MSCCA does not perform operations such as splicing and channel adjustment of features of the same level in isolation but communicates

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

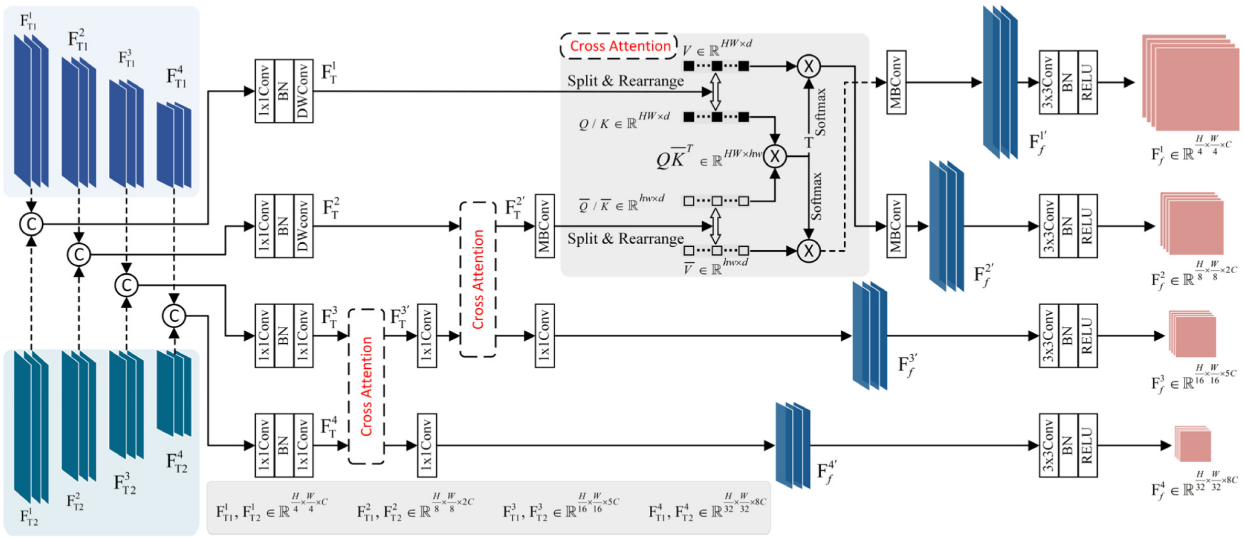


Fig. 3. Multiscale cascaded cross-attention module.

bottom-up information through cross-attention of features of different spatial resolutions and channels at each level. The cross-attention used therein is represented as follows:

$$\text{Cross-Attention}(Z, W):$$
$$\begin{cases} \overline{Q}/\overline{K}/\overline{V} = \text{Flatten}(Z) \\ Q/K/V = \text{Flatten}(W) \end{cases}$$
$$\begin{cases} Z' = \text{Reshape}\left(\text{softmax}\left(\dfrac{Q\overline{K}^T}{\sqrt{d}}\right)\overline{V}\right) \\ W' = \text{Reshape}\left(\text{softmax}\left(\dfrac{\overline{Q}K^T}{\sqrt{d}}\right)V\right) \end{cases} \tag{6}$$

where

$$\overline{Q}K^T = \left(Q\overline{K}^T\right)^{\mathrm{T}}.$$

The features $Z \in \mathbb{R}^{h\times w\times d}$ and $W \in \mathbb{R}^{H\times W\times d}$ input to cross-attention are each projected to $Q, K, V \in \mathbb{R}^{n\times d}$ and $\overline{Q}, \overline{K}, \overline{V} \in \mathbb{R}^{l\times d}$, where $l = hw, n = HW$. To reduce computational and memory consumption, both query $(\overline{Q}, Q)$ and key $(\overline{K}, K)$ of $Z$ and $W$ are shared so that the reuse of weight matrix computation can be performed by transposition operations. Cross-fusion can be realized by cross-attention module for two sets of token sequences with different lengths, which provides the basis for bottom-up hierarchical feature redistribution fusion. In the actual calculation of cross-attention, a multihead calculation is used, which is represented as follows:

$$\text{MHSA}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \tag{7}$$

where

$$\text{head}_i = \text{Attention}\left(QW_i^Q, KW_i^K, VW_i^V\right)$$

where $W_i^Q \in \mathbb{R}^{C\times d_w}$, $W_i^K \in \mathbb{R}^{C\times d_k}$, $W_i^V \in \mathbb{R}^{C\times d_v}$, $W^O \in \mathbb{R}^{hd_v\times C}$ are learnable matrices, $d_k$, $d_v$ are the hidden dimensions of the projection subspace, $h$ is the number of heads, and $C$ is the embedding dimension.

The different features extracted by the twin dual-branch encoder in the CD task need to undergo some fusion operations before they can be used by the encoder. In addition, extensive work has demonstrated that the skip connection from the encoder to the decoder reduces the information loss due to downsampling and the effect of noise introduced by the upsampling process. In order to better fuse the dual-branch features and reduce the interference of irrelevant information from shallow coarse-grained features at the end of the network during the skip connection, we adopt a bottom-up information flow based on cross-attention to fuse the deeper features with high semantic information and the shallow features with location information by cascading. As shown in Fig. 3, we first splice the dual-branch features in the channel dimension according to the corresponding hierarchy. The details are as follows:

$$F_T^i = \text{Concat}\left[F_{T1}^i, F_{T2}^i\right], \quad i = 1, 2, 3, 4. \tag{8}$$

The initial fused features obtained by splicing are then channel-adjusted, in which, we introduce depthwise separable convolution (DWConv) [50] for shallow features with large spatial resolution to collect spatial and dimensional information. Because of the high aggregation of information carried by deep features, in order to avoid introducing noise, we use $1 \times 1$ convolution for projection operation. The details are as follows:

$$\begin{aligned} F_T^i &= \text{DWConv}\left(\text{BatchNormal}\left(\text{Conv}_{1\times1}\left(F_T^i\right)\right)\right), \quad i = 1, 2 \\ F_T^i &= \text{Conv}_{1\times1}\left(\text{BatchNormal}\left(\text{Conv}_{1\times1}\left(F_T^i\right)\right)\right), \quad i = 3, 4 \\ &\text{DWConv}(x) \\ &= (x \cdot W_\text{depthwise}) \cdot W_\text{pointwise} \end{aligned} \tag{9}$$

where $W$_depthwise denotes a point-by-point convolution kernel of size $3 \times 3$, which performs independent spatial convolution operations on each channel of the input feature map to model spatial correlation. $W$_pointwise denotes a point-by-point convolution kernel of size $1 \times 1$, which performs a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

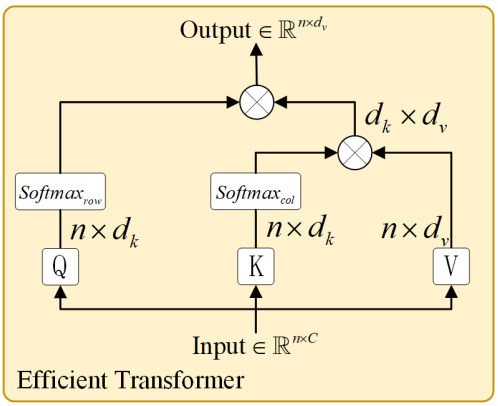


Fig. 4. Efficient transformer.

channel projection on the output feature map to model channel correlation.

Then, we use cross-attention to cascade the cross fusion of features with different spatial resolutions from deep to shallow directions. The details are as follows:

$$\begin{aligned} F_f^{4'}, F_T^{3'} &= \text{Cross-Attention}\left(F_T^4, F_T^3\right) \\ F_f^{3'}, F_T^{2'} &= \text{Cross-Attention}\left(F_T^{3'}, F_T^2\right) \\ F_f^{2'}, F_f^{1'} &= \text{Cross-Attention}\left(F_T^{2'}, F_T^1\right) \\ F_f^i &= \text{RELU}\left(\text{BatchNormal}\left(\text{Conv}_{3\times 3}\left(F_f^{i'}\right)\right)\right) \\ i &= 1, 2, 3, 4. \end{aligned} \tag{10}$$

Through MSCCA, we fused the hierarchical features extracted by the dual-branch encoder and redistributed the information so that the highly representative semantic information of the deeper features is gradually embedded into the shallower features. This further enhances the role of jump connections to increase information use. At the same time, the redundant information of the shallow features is filtered, reducing the pressure of removing noisy information at the end of the network.

### D. Efficient Transformer

As in Fig. 4, just like plain transformers, in efficient transformers [49], the input feature vector Input $\in \mathbb{R}^{n\times C}$ is still formed through three linear layers of queries $Q \in n \times d_k$, keys $K \in n \times d_k$, and values $V \in n \times d_v$. The efficient transformer, however, provides a new interpretation of attention by treating the keys as template attention maps and the queries as corresponding to the coefficients of each template, thus aggregating the value features to generate global context vectors. The following equation characterizes the efficient attention mechanism:

$$E(Q, K, V) = \text{Softmax}_{\text{row}}(Q)\left(\text{Softmax}_{\text{col}}\left(K^T\right)V\right) \tag{11}$$

where $\text{Softmax}_{\text{row}}$, $\text{Softmax}_{\text{col}}$ denote applying the SoftMax function along each row or column of the matrix, respectively.

Efficient attention provides significant improvements in computational complexity and memory consumption over the traditional dot-product attention mechanism, while maintaining the representational power of dot-product attention. Specifically, the mechanism reduces the computational complexity of the dot-product attention mechanism from $O(n^2)$ to $O(d_k d_v)$ by changing the order of the attention computation, where $d_k$ and $d_v$ are the dimensions of the keys and the values, respectively, and usually $d_k$ and $d_v$ are much smaller than $n^2$. The efficient attention mechanism can, therefore, significantly reduce the computation and memory requirements, which allows the attention mechanism to be applied to higher resolution inputs or resource constrained environments.

### E. Loss Function

We use the dice loss function [51] and the weighted cross-entropy loss function [6] together to compute the loss during network training. Considering that the RSCD task is a dense pixel-level prediction task, we describe the change detection results $\widehat{Y} \in \mathbb{R}^{H\times W}$ predicted by the network as a collection of pixel points as follows:

$$\widehat{Y} = \left\{ \widehat{y}_i \middle| i = 1, 2, 3, \ldots, H \times W \right\}. \tag{12}$$

Then, the cross-entropy loss function:

$$L_{\text{wce}} = \frac{1}{H \times W} \sum_{i=1}^{H\times W} \text{weight[class]} \cdot \left(\log\left(\frac{e^{\widehat{y}_i[\text{class}]}}{e^{\widehat{y}_i[0]} + e^{\widehat{y}_i[1]}}\right)\right). \tag{13}$$

The value range of class is {0, 1}, where 0 indicates that this pixel is predicted to change, and 1 indicates that this pixel is predicted not to change. The dice coefficient is a metric function that evaluates the similarity of the two samples, with a range of [0, 1], and larger values indicating that the samples are more similar. Denoting the set of pixels of ground truth as $Y$, the dice coefficient is defined as follows:

$$\text{DiceCoefficient} = \frac{2|Y \cap \widehat{Y}|}{|Y| + |\widehat{Y}|} \tag{14}$$

where $|\hat{Y}|, |Y|$ denote the number of pixels in the set of predicted pixels and ground truth, and $|Y \cap \hat{Y}|$ represents the number of pixels they intersect. On the basis of the dice coefficient, the Dice loss is defined as follows:

$$L_{\text{Dice}} = 1 - \text{DiceCoefficient}. \tag{15}$$

The total loss is calculated as follows:

$$L_{\text{loss}} = L_{\text{wce}} + L_{\text{dice}}. \tag{16}$$

## IV. Experiments and Results Analysis

We conducted experiments on the performance of MSCCA-Net on three publicly available change detection datasets: CDD [52], LEVIR-CD [37], and BCDD [53]. In this section, we first introduce the dataset and comparison methods used. Then, the computation of the model evaluation metrics and the experimental setup are described. Finally, the experimental results were quantitatively analyzed and visualized.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

TABLE I
THREE DATASET CONFIGURATIONS USED

| Datasets | Spatial Resolution | Size/Image | Number of Pixels in Datasets | | | Number of Images in Datasets | | |
|---|---|---|---|---|---|---|---|---|
| | | | Changed | Unchanged | Ratio | Train | Validation | Test |
| CDD | 0.03-1m/pixel | 256×256 | 134,068,750 | 914,376,178 | 1:6.82 | 10,000 | 3,000 | 3,000 |
| BCDD | 0.2m/pixel | 256×256 | 21,352,815 | 477,759,663 | 1:22.37 | 5895 | 794 | 745 |
| LEVIR-CD | 0.5m/pixel | 256×256 | 30,913,975 | 637,028,937 | 1:20.61 | 7120 | 1024 | 2048 |

### A. Dataset Settings

*1) CDD [52]:* This dataset was obtained from Google Earth containing 11 pairs of multispectral images of seasonal RS changes, which contain changes in features, such as buildings, roads, and vehicles, with a spatial resolution of 0.03–1 m/pixel. 16 000 sample pairs of 256 × 256 pixels in size were obtained from these 11 pairs of images acquired by cropping and rotating operations. This resulted in 10 000/3000/3000 image pairs for training/validation/test.

*2) LEVIR-CD [37]:* This dataset is a large building CD dataset obtained from Google Earth. It covers various types of building change characteristics, including high-rise apartments, villas, small garages, and large warehouses, etc., at a spatial resolution of 0.5 m/pixel. Most of the changes in this dataset are building growth changes. By nonoverlapping cropping operation, 10 192 sample pairs of 256 × 256 pixels in size were captured from 637 high-resolution (0.5 m/pixel) Google Earth image pairs of 1024 × 1024 pixels in size, and 7120/1024/2048 image pairs were obtained by dividing them in the ratio of 7:1:2 for training/validation/test.

*3) BCDD [53]:* This dataset is a building change detection dataset derived from aerial imagery of the Christchurch area of New Zealand after the earthquake and after reconstruction. It contains 12 796 buildings within 20.5 $km^2$ and 16 077 buildings within the same area from the 2016 data, with a spatial resolution of 0.2 m/pixel. Since there is no solution given in [53] on how to segment the samples, we segmented a pair of images with the spatial size of 32 507 × 15 354 into 256 × 256 image pairs and randomly divided them into 5895/794/745 for training/validation/test.

Table I shows the details of the three datasets mentioned above. Among them, the CDD dataset focuses on the changes in buildings, roads, and vehicles, ignoring the seasonal and luminance differences, which poses a great challenge to the judgment of pseudo changes by the change detection algorithms. LEVIR-CD focuses on the changes in dense buildings, with an average of about 50 building changes in each pair of images, and introduces the changes caused by the seasonal and luminance changes. By choosing datasets with different spatial resolutions, data volumes, and sample distributions, the robustness and generalization performance of the CD model can be verified more comprehensively and objectively.

### B. Comparison Methods

We compared MSCCA-Net with some advanced DL change detection methods. The methods used for comparison are as follows.

*1) CDNet [15]:* The method uses an inverse convolutional network with convolutional kernel size fixed at 7 and intermediate layer channels fixed at 64 for change detection. It uses a single-branch encoder with stacked image pairs as input and outputs a pixel-level classification map of structural changes.

*2) STANet [37]:* The method uses ResNet18 as the network backbone, which emphasizes the importance of spatial-temporal correlation in the CD domain and uses self-attention mechanisms to compute spatial and temporal correlations to detect changes.

*3) BIT-CD [4]:* The method combines CNNs in series with a transformer to model the spatial-temporal context locally and globally to better simulate the context. Semantically rich token sequences are then mapped to the pixel space, thereby detecting changes.

*4) ChangeFormer [8]:* The method does not use convolutional networks but uses a transformer and multilayer perceptron to build encoders and decoders for feature extraction and recovery to accomplish the change detection task.

*5) DMINet [9]:* The method is based on ResNet18 and proposes a cross-period joint-attention block to control the global feature distribution of each input by unifying self-attention and cross-attention in a single module, stimulating the coupling of information between representations within the layer while suppressing the interference of task-independent changes.

*6) ACABFNet [14]:* The method proposes a dual-branch fusion network based on axial cross-attention. This network extracts local and global information of an image through CNN branches and transformer branches and then fuses the local and global features through a two-way fusion method.

*7) WNet [10]:* The method employs a dual Siamese branching encoder, combining CNN and transformer, to extract both local and global information of RS images. Deformable convolution and attention mechanisms are introduced to enhance the model's ability to model critical regions.

*8) HANet [11]:* The method proposes a progressive foreground balanced sampling strategy to help the model learn foreground image features accurately by learning the foreground image features first and then gradually adding the background image. In addition, the HAN module is used for feature refinement. HAN module contains a parallel convolutional structure, channel attention module, and rank attention for better semantic information.

*9) PA-Former [12]:* The method uses a priori feature extractor and a priori aware transformer module for RS building change detection. The method effectively fuses a priori features and depth features to improve the detection

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

performance by extracting building a priori information and performing cross-temporal context modeling.

### C. Evaluation Metrics

In order to quantitatively compare the performance of the models, we use five evaluation metrics to synthesize the experimental results, namely precision (Pre), recall (Rec), overall accuracy (OA), intersection over union (IoU), and F1-score (F1). We use the F1 on change categories as the main quantitative indicators. Where Pre reflects the proportion of correct model predictions in the correctness sample, Rec reflects the proportion of correct model predictions in all correctness samples, and F1 is the weighted summed average of the two. OA refers to the ratio of the number of correctly categorized category likelihoods to the total number of categories. IoU reflects the correlation between the true and predicted values. Where F1 and IoU have a value range between 0 and 1. In general, the higher the value of F1, the higher the detection accuracy of the model. Each metric is calculated as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{17}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{18}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FN} + \text{TN} + \text{FP}} \tag{19}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{20}$$

$$\text{F1} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{21}$$

where TP stands for true positive, which indicates the number of change pixels correctly predicted. FP stands for false positive, which indicates the number of unchanged pixels incorrectly predicted as changes. TN stands for true negative, which indicates the number of unchanged pixels correctly predicted. FN stands for false negative, which indicates the number of change pixels incorrectly predicted as unchanged.

### D. Training Details

All experiments were performed on NVIDIA TITAN RTX (24 GB) GPUs. We implement the proposed MSCCA-Net using the PyTorch framework and the AdamW optimizer. The training is done in 200 epochs with the batch size set to 16. Exponential decay is performed after the initial learning rate is reached during the warm-up epoch. Specifically, the learning rate decays with cycles with the following equation:

$$\begin{aligned}
&\text{when } 0 \le \text{epoch} < \text{warmup_epoch} \\
&\qquad \text{lr} = \text{init_lr} \times 2.718^{\left(\frac{10 \times \text{epoch}}{\text{warm_epoch} - 1}\right)} \\
&\text{when epoch} = \text{warmup_epoch} \\
&\qquad \text{lr} = \text{init_lr} \\
&\text{when warmup_epoch} < \text{epoch} \le \text{max_epoch} \\
&\qquad \text{lr} = \text{init_lr} \times \left(1 - \frac{\text{epoch}}{\text{max_epoch}}\right)^{0.9}
\end{aligned} \tag{22}$$

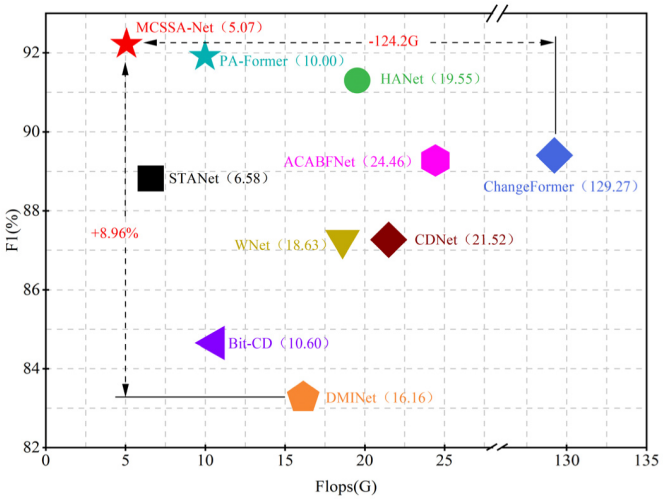


Fig. 5. Scatterplot of the number of floating-point operations for MSCCA-Net and other CD models.

where epoch denotes the current epoch number, lr is the learning rate of the current epoch, init_lr is the initial learning rate, warmup_epoch is the number of warm-up cycles, and max_epoch is the training cycle. We set the initial learning rate to 0.0001 and the warm-up period to 5. In order to ensure fairness in comparison, we retrain, validate, and test in the same hardware environment using the public code of all compared methods.

### E. Performance Comparison

The results of the quantitative comparison of each method on the CDD, LEVIR-CD, and BCDD test sets are presented in Table II. The quantitative results show that MSCCA-Net's overall performance on these datasets is significantly better than the other comparison methods. On the CDD dataset, MSCCA-Net's F1 score is in the lead. On the LEVIR-CD dataset, the F1 score of MSCCA-Net is 0.19% higher than the nearest DMINet and 0.58% higher than ACABFNet. In addition, we compare the number of floating-point operations (Flops) and the number of parameters (Params.) of MSCCA-Net with other RSCD models in Figs. 5 and 6. The data presented in Figs. 5 and 6 were all measured in the same environment using the same method (torchsummary). All F1 are taken from the results of each method on the BCDD dataset. Combined in Figs. 5 and 6 it can be seen that on the BCDD dataset, the F1 score of MSCCA-Net is 2.82% higher than ChangeFormer; however, MSCCA-Net's floating-point operations and parameter quantity only account for 3.9% and 13.2% of ChangeFormer's. In addition, MSCCA-Net has a relatively small number of parameters in the model that employ the transformer architecture. This indicates that MSCCA-Net balances CD performance and computational cost well.

In order to compare the detection effect of each method more intuitively, we use multiple colors to indicate the detection results in Fig. 7: TP (white), TN (black), FP (red), and FN (green).

Overall, models that combine CNNs with transformers, such as PA-Former, WNet, and MSCCA-Net, perform better in terms of detection effectiveness. The performance of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON RESULTS ON THE THREE CD TEST SETS. BOLD INDICATES HIGHEST SCORE. ALL VALUES ARE PRESENTED AS PERCENTAGES (%)

| Model | CDD<br>Pre. / Rec. / F1-score / IoU / OA | LEVIR-CD<br>Pre. / Rec. / F1-score / IoU / OA | BCDD<br>Pre. / Rec. / F1-score / IoU / OA |
|---|---|---|---|
| CDNet | 91.93 / 84.87 / 88.26 / 78.98 / 97.34 | 88.38 / 85.08 / 86.70 / 76.52 / 98.67 | 90.96 / 83.86 / 87.26 / 77.41 / 98.85 |
| STANet | 88.97 / 94.31 / 91.56 / 84.44 / 97.95 | 80.99 / **91.21** / 85.79 / 75.12 / 98.46 | 89.01 / 88.63 / 88.82 / 79.89 / 98.95 |
| Bit-CD | 95.86 / 94.59 / 95.22 / 90.88 / 98.88 | **91.95** / 88.57 / 90.23 / 82.19 / 99.02 | 85.62 / 83.68 / 84.64 / 73.37 / 98.68 |
| ChangeFormer | 95.27 / 93.82 / 94.54 / 89.65 / 98.72 | 91.53 / 88.86 / 90.17 / 82.10 / 99.01 | 93.44 / 85.70 / 89.40 / 80.84 / 99.12 |
| DMINet | 96.53 / 96.65 / 96.59 / 93.41 / 99.19 | 92.33 / 89.40 / 90.84 / 83.23 / 99.08 | 87.19 / 79.68 / 83.26 / 71.33 / 98.61 |
| ACABFNet | 96.59 / 96.69 / 96.64 / **96.30** / 99.20 | 91.38 / 89.54 / 90.45 / **90.78** / 99.03 | 87.47 / **91.15** / 89.27 / **89.77** / 98.97 |
| WNet | 97.17 / 97.09 / 97.13 / 94.42 / 99.32 | 91.31 / 89.41 / 90.35 / 82.40 / 99.02 | 88.80 / 85.77 / 87.26 / 77.40 / 98.82 |
| HANet | 92.44 / 84.68 / 88.39 / 79.19 / 97.12 | 89.56 / 87.13 / 88.33 / 79.10 / 98.83 | 92.73 / 89.91 / 91.30 / 83.99 / 99.20 |
| PA-Former | 94.67 / 95.55 / 95.11 / 90.68 / 98.85 | 89.00 / 91.20 / 90.09 / 81.97 / 99.00 | 90.70 / 93.16 / 91.91 / 85.04 / 99.25 |
| MSCCA-Net | **97.12** / **97.28** / **97.20** / 94.56 / **99.35** | 91.55 / 90.50 / **91.03** / 83.53 / **99.09** | **93.63** / 90.84 / **92.22** / 85.56 / **99.28** |

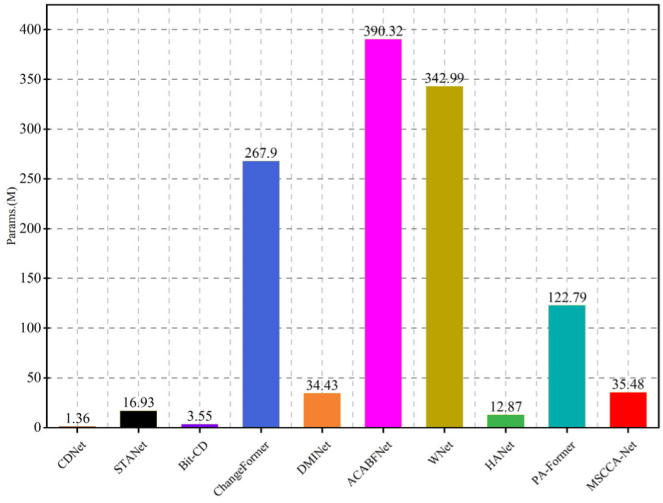


Fig. 6. Bar chart of the number of parameters for MSCCA-Net and other CD models.

MSCCA-Net on road change detection is outstanding, which can reduce the influence of irrelevant changes in complex scenes and accurately recognize road changes. As shown in Fig. 7(a) and (c), it is not easy to accurately detect road changes due to the influence of vegetation and snow in the area to be examined. It can be seen that CDNet and HANet have a more serious miss-detection of road changes (green color). STANet is more affected by uncorrelated changes and has more mis-detections of road edges (red color). For the changes in complex buildings affected by lighting conditions [e.g., Fig. 7(b)], MSCCA-Net and DMINet can well exclude the influence of shadows and irregular small targets and more accurately detect the changes in building edges. For single building changes [e.g., Fig. 7(d), (g), (h), and (l)], MSCCA-Net and these comparative methods are basically able to detect them relatively well; however, for isolated small target buildings, some methods, such as CDNet, STANet, ACABFNet, and HANet will have wrong detection and omission phenomenon, while MSCCA-Net is able to detect a single building change more perfectly.

When facing the detection task of dense buildings [e.g., Fig. 7(e), (i), (j), and (k)], MSCCA-Net does not ignore the small targets at the edge of the region to be detected. As shown in Fig. 7(j) and (k), recently proposed methods such as Bit-CD, ChangeFormer, and ACABFNet all suffer from the leakage of small targets at the edges. Among them, ChangeFormer's detection of the boundary is also more complete due to the transformer's ability to capture long-range information, but the lack of inductive bias leads to the lack of local information supplementation, and the detailed information of the detected objects is lost, resulting in the leakage of fine-grained changes and the wrong detection of small irrelevant changes. Bit-CD and ACABFNet, although they both to some extent combine global and local information to a certain extent, neither of them considers the problem of fusing and fully using the two-branch hierarchical features in the process of making skip connections. HANet has integrated multiscale hierarchical features, but the use of multiscale information is still not adequate; moreover, during the skip connection process, the coarse-grained information from the shallow layers of the encoder may introduce noise, which can affect the final results. The difference is that both MSCCA-Net and DMINet process the hierarchical information in the process of skip connection accordingly, so both of them perform better in the detection of small targets. MSCCA-Net, on the other hand, has a more thorough fusion of hierarchical information through the MSCCA module, which is able to reduce the influence of noise and irrelevant changes in these application scenarios and detect semantic changes of interest more accurately.

Because of the fact that false positive rate (FPR) focuses on negative cases and true positive rate (TPR) focuses on positive cases in the receiver operating characteristic (ROC) curve, in the case of sample imbalance, the ROC curve can more objectively assess the model performance, where the implicit goal of area under the curve (AUC) is to handle cases where the sample distribution is skewed and not over-fit individual classes. In addition, in the CD domain, we focus more on how much changes are detected, i.e., more focus on positive cases. So, we also plotted precision-recall (PR) curves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

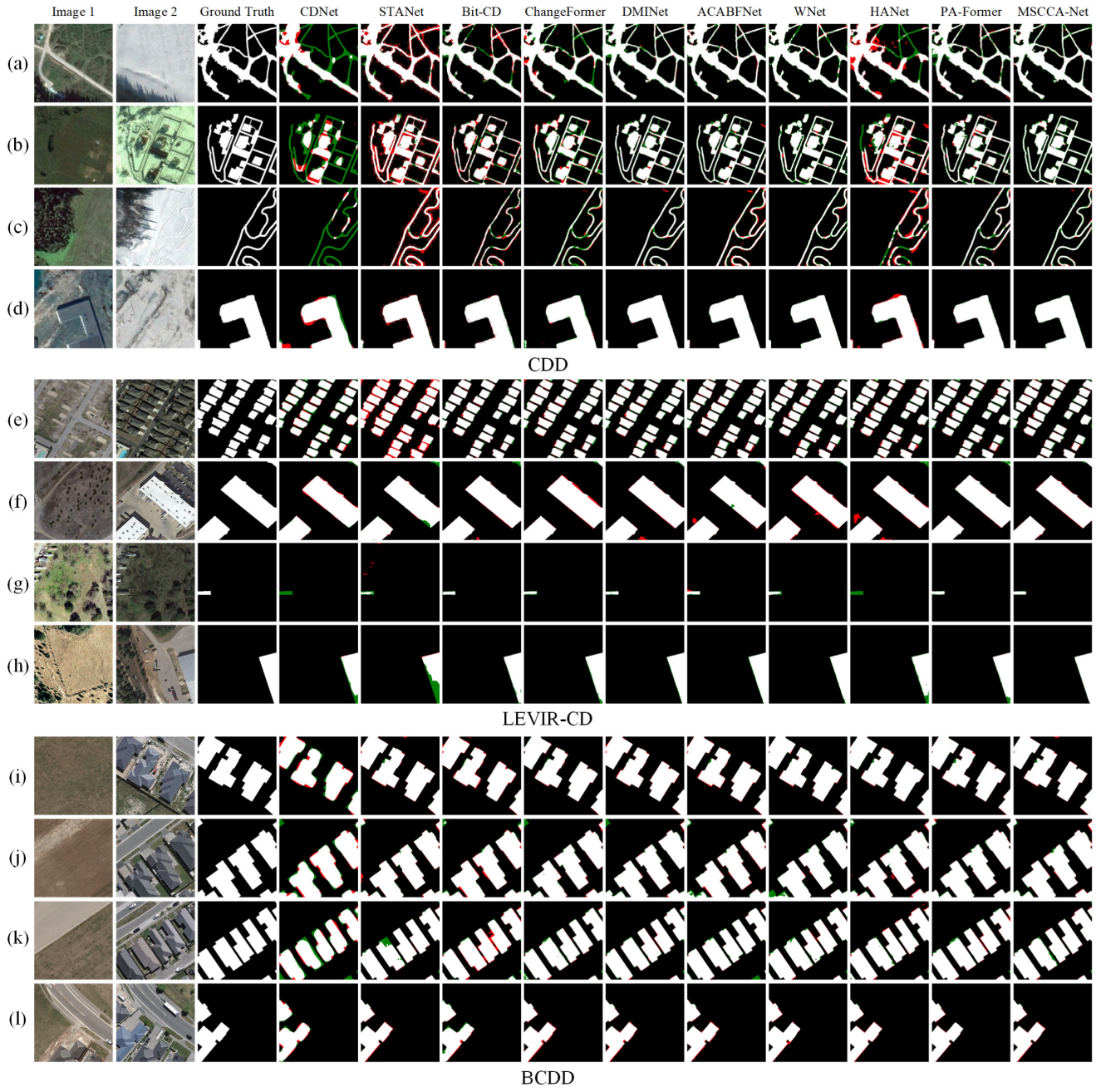


Fig. 7. Visualization results of each method on different test sets. Rows (a)–(l) represent the prediction results of each method, respectively.

to compare the ability of each model to detect positive samples in the case of sample imbalance. When dealing with highly skewed datasets, the PR curves demonstrate more information about model performance and can expose differences between models that are not evident in ROC space [54]. In summary, PR and ROC curves provide a more comprehensive assessment of the overall performance of the model than F1. Based on the results before binarization (sigmoid operation in the last layer of the network), we plotted the ROC and PR curves of each method on the BCDD dataset. Some works in the CD field have also used the ROC curve and PR curve approach to evaluate models [55], [56]. The higher the average precision (AP) and AUC values, the better the model performance and the better the generalization ability. As shown in Fig. 8, the PR curve of MSCCA-Net has the largest area under the curve with an AP of 0.9653, and the ROC curve of MSCCA-Net, compared with the comparison methods, deviates the most from the 45° diagonal line, with an AUC of 0.9955. The ROC curves and PR curves again show that the overall performance of the proposed MSCCA-Net outperforms other state-of-the-art methods.

## V. Discussion

### A. Ablation Analysis

We removed the main MSCCA-Net components one by one and verified the effectiveness of each key component

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

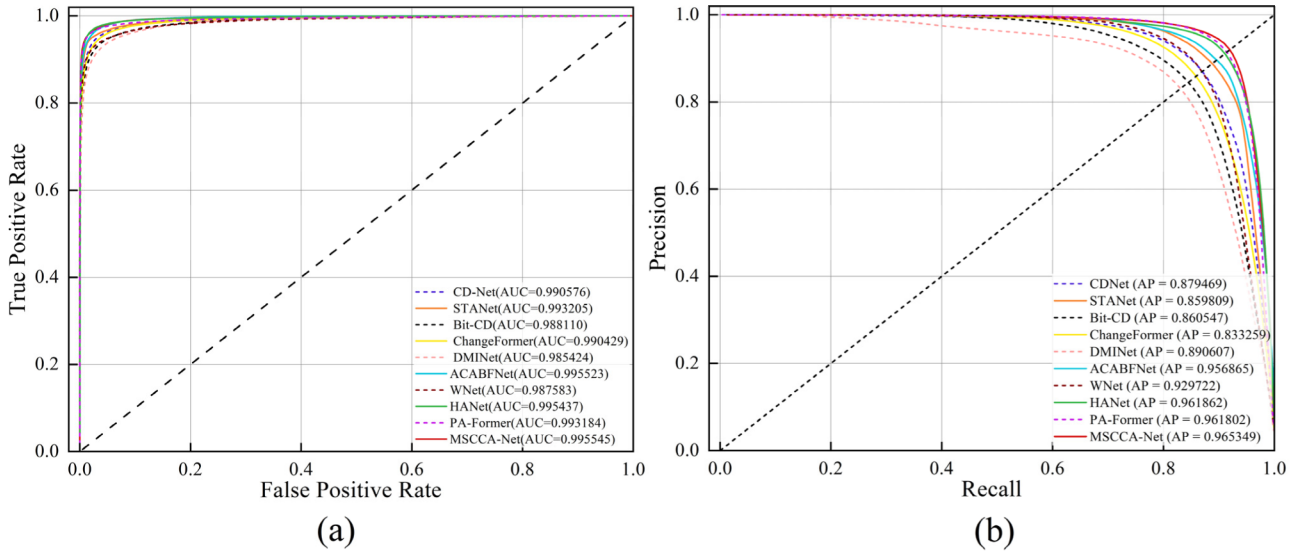


Fig. 8. PR and ROC curves for MSCCA-Net and other CD models on the BCDD dataset. (a) PR curve plotted for each model. (b) ROC curve plotted for each model.

used in MSCCA-Net on the three public datasets used. The experimental results are shown in Table III, where "w" and "w/o" are abbreviations for "with" and "without" and "ETransformer" is an abbreviation for efficient transformer. In order to further validate the effectiveness of the multiscale cascading cross-attention module, we also replace the MSCCA module with DBridge [57] for redistribution of hierarchical feature information. In addition, Top-down indicates that the cascade fusion direction of MSCCA is set to go from shallow features to deep features, and this experiment is to verify the effectiveness of the fusion direction.

We can see from Table III that when we remove the MFF and directly perform the concatenate operation to realize multibranch fusion with features extracted from different convolutional kernels, the performance of the model decreases on all three datasets. The F1 metrics are reduced by 1.24%, 0.35%, and 2.62%, respectively. This is due to the fact that the fusion of different fine-grained information realized by simple concatenation with $1 \times 1$ convolution is relatively coarse, and the network cannot discriminate the importance of each branch feature well. MFF globally encodes spatial information through horizontal and vertical biaxial adaptive pooling of spliced branch features, and the generated weight matrix reweights multibranch features to improve the effectiveness of the fusion of different fine-grained information.

We validate the effectiveness of the proposed MSCCA module by removing the MSCCA module and replacing the MSCCA module with the DBridge [57] module. From Table III, we can see that the F1 metrics of the model on the three datasets are reduced by 2.75%, 0.58%, and 1.96%, respectively, when the MSCCA module is removed, i.e., when skip connections are performed directly. When we use the DBridge [57] module to redistribute the hierarchical features during the skip connection process, the F1 metrics are improved by 1.77%, 0.1%, and −0.12%, respectively, compared to the direct skip connection. The F1 metrics, however, decreased by 0.98%, 0.48%, and 2.08%, respectively, when compared with the processing of skip features using MSCCA. These three sets of data first illustrate the inadequacy of using the information in the process of skip connection and the noise problem that may be brought about by the shallow features through the skip connection. Second, it illustrates that the cascade fusion of features in the skip connection process using the MSCCA module is a more effective modeling strategy than DBridge. This is due to the fact that DBridge first splices the hierarchical features in the channel dimension and then directly performs the separation operation after processing through the channel attention transformer layer and the token attention transformer layer. Although this can play a certain degree of redistribution of hierarchical features, the original hierarchical features change a lot, and the direct separation of the processed fusion features into four hierarchical features cannot guarantee the reasonableness of feature distribution. MSCCA's cross-fusion of adjacent hierarchical features at each level, therefore, maintains most of the original features at the corresponding level and interacts with the adjacent levels, thus improving the use of deep information.

We used an efficient transformer in the shallow layer of the encoder for initial global information extraction. This is due to the fact that the shallow layer is coarser and does not cause loss of local information, while the deeper layer of the encoder uses a stacked convolutional kernel to retain the local information. After the features obtained from up-sampling and skip connected features are fused, the decoder branch uses an efficient transformer to filter and recover the fused change information. As can be seen in Table III, replacing the efficient transformer block with convolution resulted in a decrease in the F1 value of the model by 0.89%, 0.18%, and 1.40% on the three datasets, respectively.

We know that shallow features carry rich location information, and deep features have more semantic information. Proper fusion of the two can improve the CD performance of the model. Our proposed MSCCA module is used to perform a cascade-by-cascade cross-fusion operation on neighboring

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

TABLE III
ABLATION EXPERIMENTS RESULTS ON CDD, LEVIR-CD, AND BCDD. BOLD INDICATES HIGHEST SCORE. ALL THE SCORES ARE DESCRIBED IN PERCENTAGES (%)

| Model | CDD<br>Pre. / Rec. / F1-score / IoU / OA | LEVIR-CD<br>Pre. / Rec. / F1-score / IoU / OA | BCDD<br>Pre. / Rec. / F1-score / IoU / OA |
|---|---|---|---|
| w/o MFF | 96.10 / 95.82 / 95.96 / 92.24 / 99.04 | 91.57 / 89.80 / 90.68 / 82.95 / 99.06 | 92.60 / 86.80 / 89.60 / 81.17 / 99.06 |
| w/o bridge | 95.23 / 93.68 / 94.45 / 89.49 / 98.70 | 90.62 / 90.27 / 90.45 / 82.57 / 99.02 | 92.69 / 87.95 / 90.26 / 82.24 / 99.11 |
| w DBridge[57] | 96.43 / 96.01 / 96.22 / 92.72 / 99.11 | 91.08 / 90.03 / 90.55 / 82.74 / 99.04 | 91.02 / 89.28 / 90.14 / 82.05 / 99.08 |
| w ETransformer | 96.36 / 96.25 / 96.31 / 92.88 / 99.13 | **92.22** / 89.53 / 90.85 / 83.23 / 99.08 | 93.61 / 88.19 / 90.82 / 83.18 / 99.16 |
| Top-down | 96.26 / 96.17 / 96.22 / 92.71 / 99.10 | 91.83 / 89.84 / 90.83 / 83.20 / 99.07 | **94.53** / 86.81 / 90.50 / 82.66 / 99.15 |
| MSCCA-Net | **97.12** / **97.28** / **97.20** / **94.56** / **99.35** | 91.55 / **90.50** / **91.03** / **83.53** / **99.09** | 93.63 / **90.84** / **92.22** / **85.56** / **99.28** |

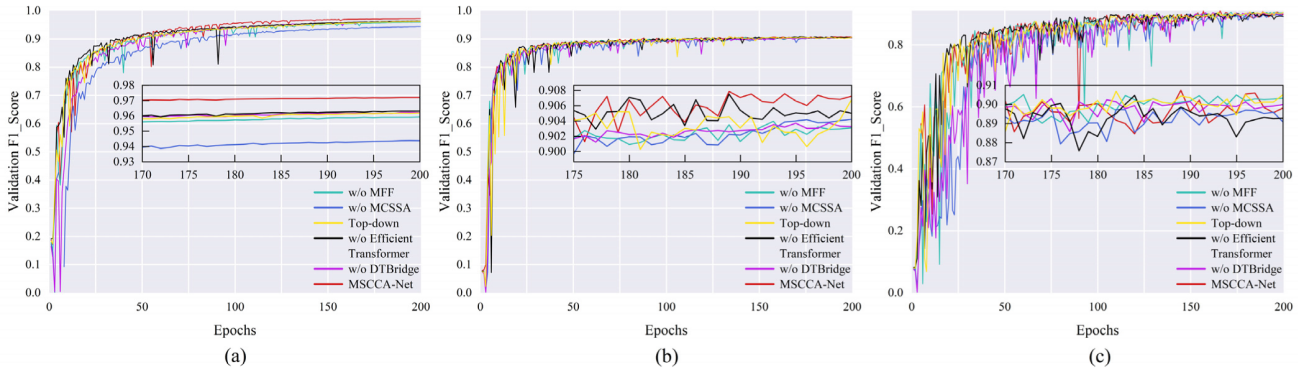


Fig. 9. F1-Score line charts, small window to zoomed-in view on the end of the line chart. (a) F1-Score line charts on the CDD dataset, (b) F1-Score line charts on the LEVIR-CD dataset, and (c) F1-Score line charts on the BCDD dataset.

hierarchical features. This will exist in two cascading directions. One is the bottom-up, i.e., cascade fusion direction from deep features to shallow features. The other is top-down, i.e., cascade fusion direction from shallow features to deep features. First of all, the flow of information should be from the direction of high effective information density to the place of low effective information density so as to improve the use of effective information in the decoding process layer by layer. Second, when fusing features from the top-down direction, the noise in the shallow layer will affect the highly aggregated feature information in the deep layer step by step; therefore, we construct the MSCCA module based on the bottom-up direction. Table III shows that when the information flow direction of MSCCA is top-down, although it can also play a certain role, the F1 index decreases by 0.98%, 0.2%, and 1.72% on the three datasets, respectively.

As shown in Fig. 9, in order to visualize the role of each module more, on the validation set of the three datasets used, we plotted line graphs of the F1 metrics for each ablation experiment. From the line graphs, we can see the overall impact of the proposed modules and strategies on the model during the training process. The MSCCA-Net training process clearly converges faster with lesser fluctuations than the other ablation models in the setup.

### *B. Visualization and Analysis of the Mechanism of Action of the MSCCA Module*

In order to illustrate the information redistribution role played by the MSCCA module among the layered features, we calculated the semantic similarity of the extracted layered features before and after going through the MSCCA module using cosine similarity, respectively. In this case, due to the large spatial resolution of the layered features, we uniformly downsampled the layered features to a size of $4 \times 4$ using the maximum pooling operation for ease of observation. This is equivalent to computing the semantic similarity between different spatial regions of the features. We visualize the cosine similarity using heatmaps after taking its absolute value. Row (a) of Fig. 10 shows the hierarchical features that are initially fused but have not gone through the MSCCA module. From left to right, we can observe that the semantic similarity between different regions of the feature decreases and then increases. This is due to the fact that shallow features have more noise information, and the network first performs a coarse extraction of feature information, which leads to a decrease in semantic similarity. As the depth of the network increases and the spatial resolution of the features decreases, the extracted feature map information is highly aggregated, which is manifested in the heat map as a higher similarity of semantic information between different regions of the features. Row (b) of Fig. 10 shows the hierarchical features after the MSCCA module. Combined with row (a) from right to left, it corresponds to the bottom-up information flow process of the MSCCA module. We can see that the semantic similarity between different regions of the shallow features increases. This indicates that the feature information flows from the high information density to the low density. The deeper semantic information flows upward in a step-by-step manner, increasing the use through skip connections.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

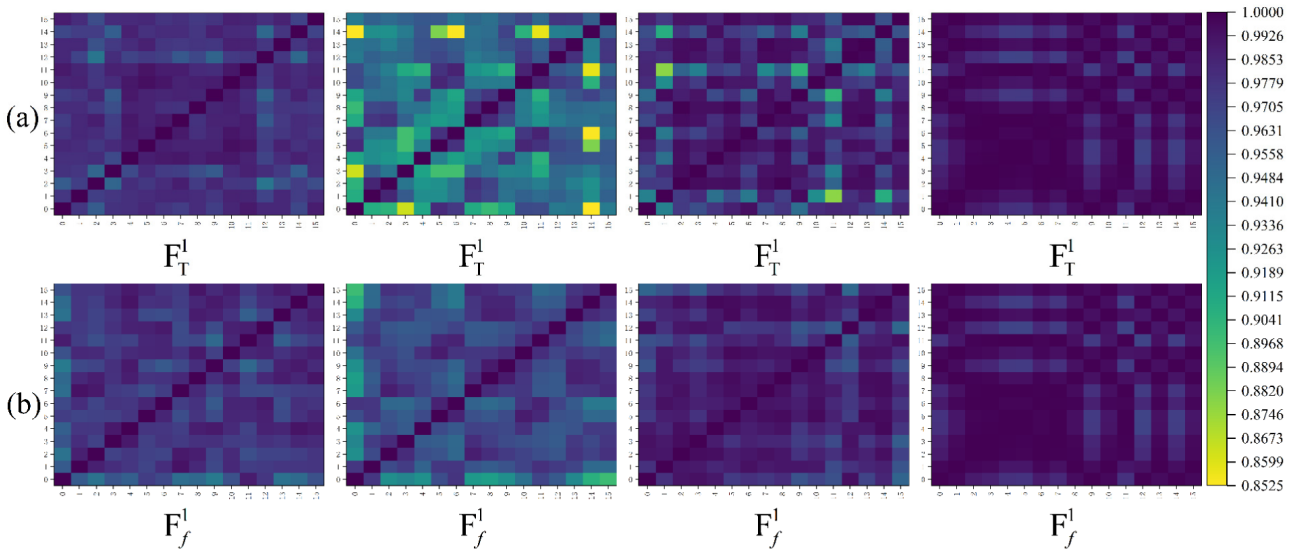


Fig. 10. Regional semantic correlation of hierarchical features. (a) Stratified features initially fused but not characterized by MSCCA. (b) After the hierarchical characterization of the MSCCA module.

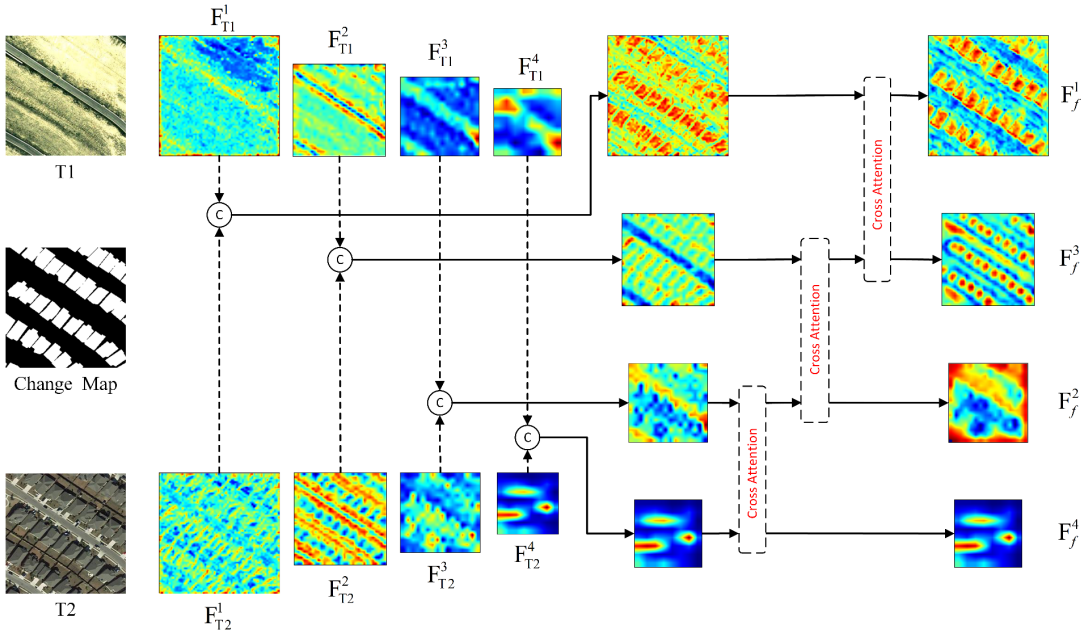


Fig. 11. Bitemporal RS image of LEVIR-CD is used as an example for MSCCA feature visualization.

In order to further illustrate the mechanism of the MSCCA module, in Fig. 11, we visualize the hierarchical feature maps before and after going through the MSCCA module. For the convenience of display, we use linear interpolation to amplify deep-level features with lower spatial resolution to 256 × 256. In the visualization, red indicates the region where the model pays more attention, and blue indicates the region where the model pays less attention. From the visualization, we can see that the dual-branch encoder first identifies the semantic objects in each image, and after the initial fusion, the model starts to focus on the parts that may have changed. In addition, we can clearly see from the shallow hierarchical features that after cascading cross-fusion, the regions that may have changed receive more attention from the network.

Combining the semantic similarity heatmap and the hierarchical feature visualization images, we can conclude that MSCCA enhances the role of the skip connection mechanism in the recovery of change features in the RSCD domain by redistributing the information from the high-density direction to the low-density direction in a stepwise manner.

## VI. Conclusion

Aiming at the problem of misdetection and omission of small targets caused by noise and information not being fully used, which can be introduced by hierarchical features connected by skip connection in the RSCD domain, we propose an end-to-end CD network based on transformer and CNN. The feature extraction backbone of this network uses a large kernel

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

convolution formed by stacking combined with an efficient transformer to achieve local and global feature extraction and fusion. Among them, in order to solve the problems of noise introduced by skip connections and the underuse of hierarchical information, we propose a multiscale cascade cross-fertilization attention, which realizes the redistribution of feature information through bottom-up information flow pairs in the fusion process, increasing the use of deep semantic information while avoiding the noise effect brought by shallow skip connections. We conducted extensive experiments on three publicly available datasets, i.e., CDD, LEVIR-CD, and BCDD. The experimental quantitative results and visualization results show that the overall performance of our proposed NSCCA-Net outperforms other comparative methods. We also provide a detailed demonstration of the mechanism of action of the MSCCA module, illustrating the necessity of reorganizing the hierarchical features before going through skip connections. Future work will continue to focus on further increasing the interaction between hierarchical features to enhance the adjunctive role of the skip connection mechanism in RSCD tasks.

## REFERENCES


[1] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[2] P. Washaya, T. Balz, and B. Mohamadi, "Coherence change-detection with Sentinel-1 for natural and anthropogenic disaster monitoring in urban areas," *Remote Sens.*, vol. 10, no. 7, p. 1026, Jun. 2018.

[3] M. Malmir, M. M. K. Zarkesh, S. M. Monavari, S. A. Jozi, and E. Sharifi, "Urban development change detection based on multi-temporal satellite images as a fast tracking approach—A case study of Ahwaz County, Southwestern Iran," *Environ. Monitor. Assessment*, vol. 187, no. 3, p. 108, Mar. 2015.

[4] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[5] Q. Li, R. Zhong, X. Du, and Y. Du, "TransUNetCD: A hybrid transformer network for change detection in optical remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622519.

[6] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[7] C. Zhang, L. Wang, S. Cheng, and Y. Li, "SwinSUNet: Pure transformer network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224713.

[8] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," 2022, *arXiv:2201.01293*.

[9] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[10] X. Tang, T. Zhang, J. Ma, X. Zhang, F. Liu, and L. Jiao, "WNet: W-shaped hierarchical network for remote-sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615814.

[11] C. Han, C. Wu, H. Guo, M. Hu, and H. Chen, "HANet: A hierarchical attention network for change detection with bitemporal very-high-resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 3867–3878, 2023.

[12] M. Liu, Q. Shi, Z. Chai, and J. Li, "PA-Former: Learning prior-aware transformer for remote sensing building change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[13] T. Lei et al., "Ultralightweight spatial–spectral feature cooperation network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600417.

[14] L. Song, M. Xia, L. Weng, H. Lin, M. Qian, and B. Chen, "Axial cross attention meets CNN: Bibranch fusion network for change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 32–43, 2023.

[15] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Auto. Robots*, vol. 42, no. 7, pp. 1301–1322, 2018.

[16] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[17] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," 2015, *arXiv:1505.04597*.

[18] Y. Zhao, P. Chen, Z. Chen, Y. Bai, Z. Zhao, and X. Yang, "A triple-stream network with cross-stage feature fusion for high-resolution image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4402114.

[19] M. Zhang, W. Li, Y. Zhang, R. Tao, and Q. Du, "Hyperspectral and LiDAR data classification based on structural optimization transmission," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3153–3164, May 2023.

[20] J. Wang, W. Li, Y. Gao, M. Zhang, R. Tao, and Q. Du, "Hyperspectral and SAR image classification via multiscale interactive fusion network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 10823–10837, Dec. 2023, doi: 10.1109/TNNLS.2022.3171572.

[21] X. Zhang, S. Cheng, L. Wang, and H. Li, "Asymmetric cross-attention hierarchical network based on CNN and transformer for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2000415.

[22] E. J. M. Rignot and J. J. van Zyl, "Change detection techniques for ERS-1 SAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 31, no. 4, pp. 896–906, Jul. 1993.

[23] R. A. Weismiller, S. J. Kristof, D. K. Scholz, P. E. Anuta, and S. A. Momin, "Change detection in coastal zone environments," *Photogramm. Eng. Remote Sens.*, vol. 43, pp. 1533–1539, Dec. 1977.

[24] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," in *Proc. LARS Sym.*, 1980, pp. 1–12.

[25] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and $k$-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

[26] W. Yan, S. Shi, L. Pan, G. Zhang, and L. Wang, "Unsupervised change detection in SAR images based on frequency difference and a modified fuzzy c-means clustering," *Int. J. Remote Sens.*, vol. 39, no. 10, pp. 3055–3075, May 2018.

[27] T. Kasetkasem and P. K. Varshney, "An image change detection algorithm based on Markov random field models," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 8, pp. 1815–1823, Aug. 2002.

[28] K. Zong, A. Sowmya, and J. Trinder, "Building change detection from remotely sensed images based on spatial domain analysis and Markov random field," *J. Appl. Remote Sens.*, vol. 13, no. 2, p. 1, May 13, 2019.

[29] J. Im and J. Jensen, "A change detection model based on neighborhood correlation image analysis and decision tree classification," *Remote Sens. Environ.*, vol. 99, no. 3, pp. 326–340, Nov. 2005.

[30] H. Nemmour and Y. Chibani, "Multiple support vector machines for land cover change detection: An application for mapping urban extensions," *ISPRS J. Photogramm. Remote Sens.*, vol. 61, no. 2, pp. 125–133, Nov. 2006.

[31] J. J. Gapper, H. El-Askary, E. Linstead, and T. Piechota, "Coral reef change detection in remote Pacific Islands using support vector machine classifiers," *Remote Sens.*, vol. 11, no. 13, p. 1525, Jun. 2019.

[32] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Proc. 4th Int. Workshop Deep Learn. Med. Image Anal. Multimodal Learn. Clin. Decis. Support*, vol. 11045, Sep. 2018, pp. 3–11.

[33] M. Zhang, G. Xu, K. Chen, M. Yan, and X. Sun, "Triplet-based semantic relation learning for aerial remote sensing image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 266–270, Feb. 2019.

[34] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[35] J. Zheng, Y. Feng, C. Bai, and J. Zhang, "Hyperspectral image classification using mixed convolutions and covariance pooling," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 522–534, Jan. 2021.

[36] G. Alimjan, Y. Jiaermuhamaiti, H. Jumahong, S. Zhu, and P. Nurmamat, "An image change detection algorithm based on multi-feature self-attention fusion mechanism UNet network," *Int. J. Pattern Recognit. Artif. Intell.*, vol. 35, no. 14, Nov. 2021.

[37] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.

[38] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.
[39] Z. Tian, J. Yi, Y. Bai, J. Tao, S. Zhang, and Z. Wen, "Synchronous transformers for end-to-end speech recognition," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 7884–7888.
[40] A. Dosovitskiy et al., "An image is worth 16 × 16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[41] A. Vaswani et al., "Attention is all you need," 2017, *arXiv:1706.03762*.
[42] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[43] G. Wang, B. Li, T. Zhang, and S. Zhang, "A network combining a transformer and a convolutional neural network for remote sensing image change detection," *Remote Sens.*, vol. 14, no. 9, p. 2228, May 2022.
[44] F. Song, S. Zhang, T. Lei, Y. Song, and Z. Peng, "MSTDSNet-CD: Multiscale Swin Transformer and deeply supervised network for change detection of the fast-growing urban regions," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[45] H. Wang, S. Cheng, Y. Li, and A. Du, "Lightweight remote-sensing image super-resolution via attention-based multilevel feature fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 6508505.
[46] T. Lei, Y. Zhang, Z. Lv, S. Li, S. Liu, and A. K. Nandi, "Landslide inventory mapping from bitemporal images using deep convolutional neural networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 6, pp. 982–986, Jun. 2019, doi: 10.1109/LGRS.2018.2889307.
[47] Z. Fu, J. Li, L. Ren, and Z. Chen, "SLDDNet: Stagewise short and long distance dependency network for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 3000319.
[48] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.
[49] S. Zhuoran, Z. Mingyuan, Z. Haiyu, Y. Shuai, and L. Hongsheng, "Efficient attention: Attention with linear complexities," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 3530–3538.
[50] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1800–1807.
[51] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.
[52] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 2, pp. 565–571, May 2018.
[53] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2018.
[54] J. Davis and M. Goadrich, "The relationship between precision-recall and ROC curves," in *Proc. 23rd Int. Conf. Mach. Learn.*, 2006, pp. 233–240.
[55] Q. Ding, Z. Shao, X. Huang, and O. Altan, "DSA-Net: A novel deeply supervised attention-guided network for building change detection in high-resolution remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102591.
[56] L. Lei, Y. Sun, and G. Kuang, "Adaptive local structure consistency-based heterogeneous remote sensing change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[57] R. Azad, Y. Jia, E. K. Aghdam, J. Cohen-Adad, and D. Merhof, "Enhancing medical image segmentation with TransCeption: A multi-scale feature fusion approach," 2023, *arXiv:2301.10847*.

**Xiaofeng Zhang** received the B.E. degree from the Henan University of Technology, Zhengzhou, China, in 2021. He is currently pursuing the master's degree with the School of Computer Science and Technology, Xinjiang University, Ürümqi, China.

His research interests include computer vision and remote sensing image change detection.

**Liejun Wang** received the Ph.D. degree from the School of Information and Communication Engineering, Xi'an Jiaotong University, Xi'an, China, in 2012.

He is currently a Professor with the School of Computer Science and Technology, Xinjiang University, Ürümqi, China. His research interests include computer vision and natural language processing.

**Shuli Cheng** received the Ph.D. degree from the School of Computer Science and Technology, Xinjiang University, Ürümqi, China, in 2021.

He is currently working with the School of Computer Science and Technology, Xinjiang University. His research interests include computer vision, multimodal processing, and remote sensing image change detection.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 20:29:33 UTC from IEEE Xplore. Restrictions apply.